	AIR UNIVERSITY
	DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
	EXPERIMENT NO 13

Lab Title: Open Ended lab

Student Name: Muhammad Waqas **Reg. No:** 240756

Objective: To design a generic circuit that implements both adder and subtractor circuit.

To design BCD to Seven Segment Display decoder using logic gates

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LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks: _____

Date: _____ Signature: _____

EXPERIMENT 13

Open Ended Lab

Objectives:

- To design a generic circuit that implements both adder and subtractor circuit.
- To design BCD to Seven Segment Display decoder using logic gates.

Equipment required:

- ICs as required
- Cables
- Digital Electronics Trainer

Task 1:

Design a circuit that implements a half adder as well as a half subtractor. Use a selection switch (S_W) to switch between adder and subtractor. The circuit should perform addition when SW = 0 and subtraction when SW = 1.

Steps:

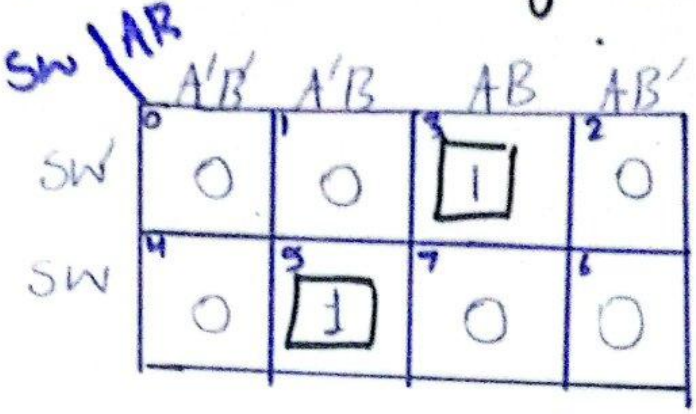
1. Complete the Truth Table according to the given requirements.
2. Write the logic equations (involving the switch).
3. Draw the logic diagram of your design.

Truth Table:

Truth Table				
S_W	**A**	**B**	**C/B**	**S/D**
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	0
1	0	1	1	1
1	1	0	0	1
1	1	1	0	0

Logic Equations for Carry and Borrow:	Logic Equations for Sum and Differenec:

Logic Equation for carry and Borrow

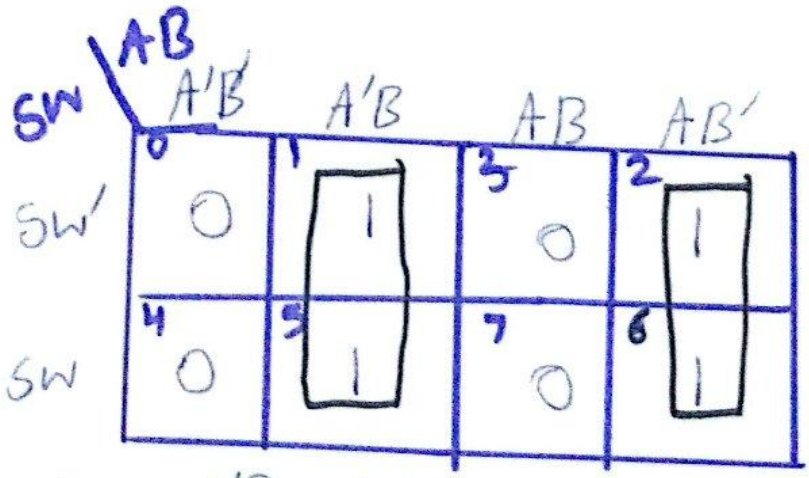


$C/B = SW'AB + SWA'B$

$= B(SW'A + SWA')$

$C/B = B(SW \oplus A)$

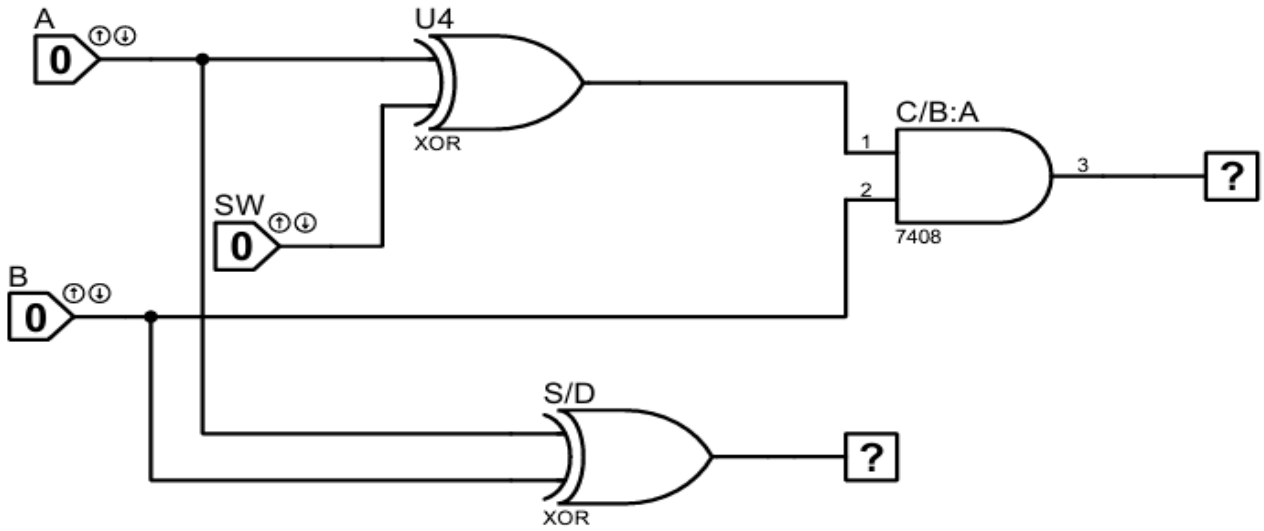
Logic Equation for Sum and Differenc o



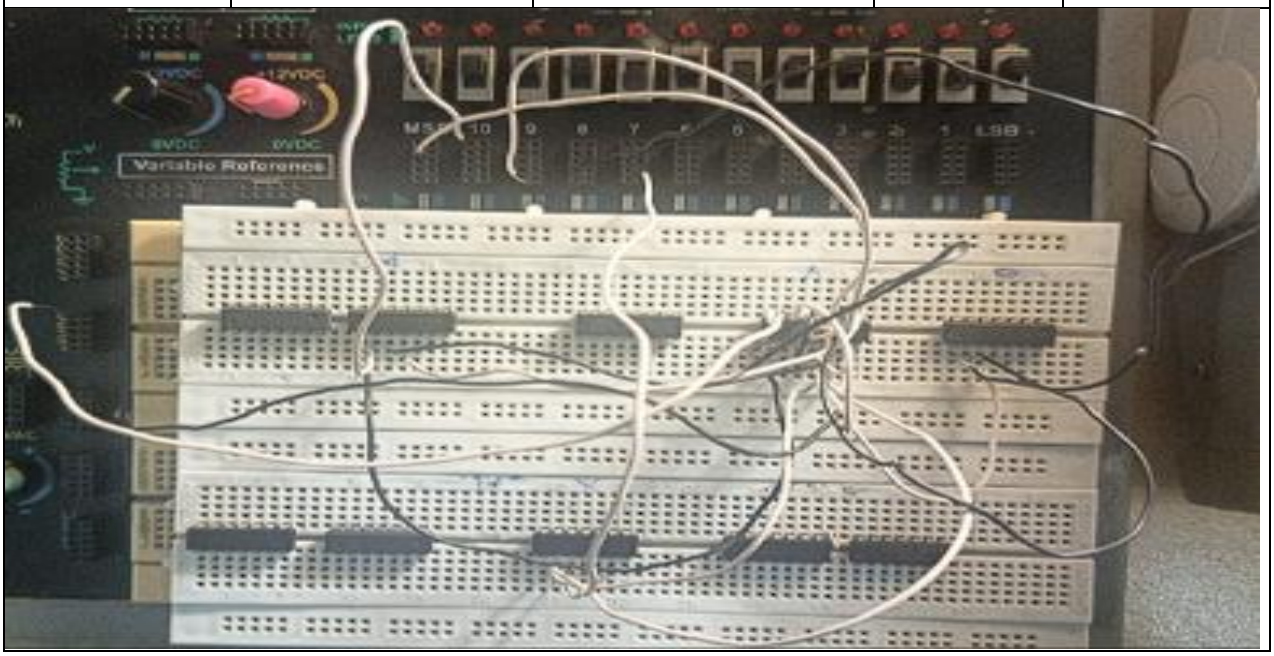
$S/D = A'B + AB'$

$= A \oplus B$

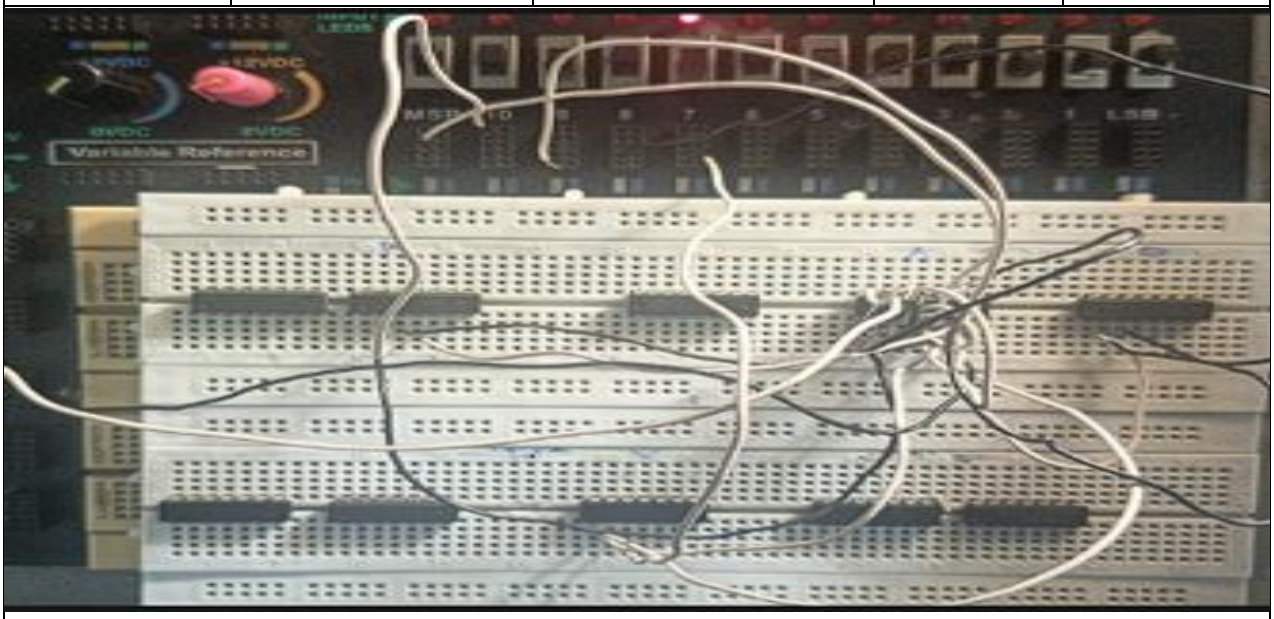
Logic Diagram:



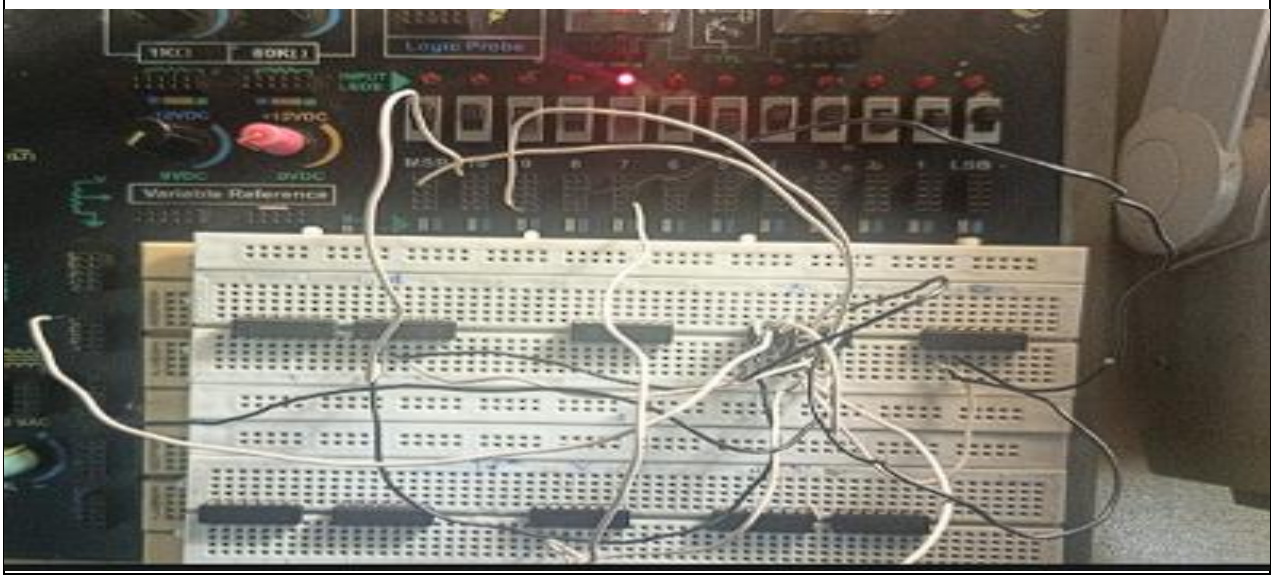
Truth Table(HALF ADDER)				
Sw	**A**	**B**	**C/B**	**S/D**
0	0	0	0	0



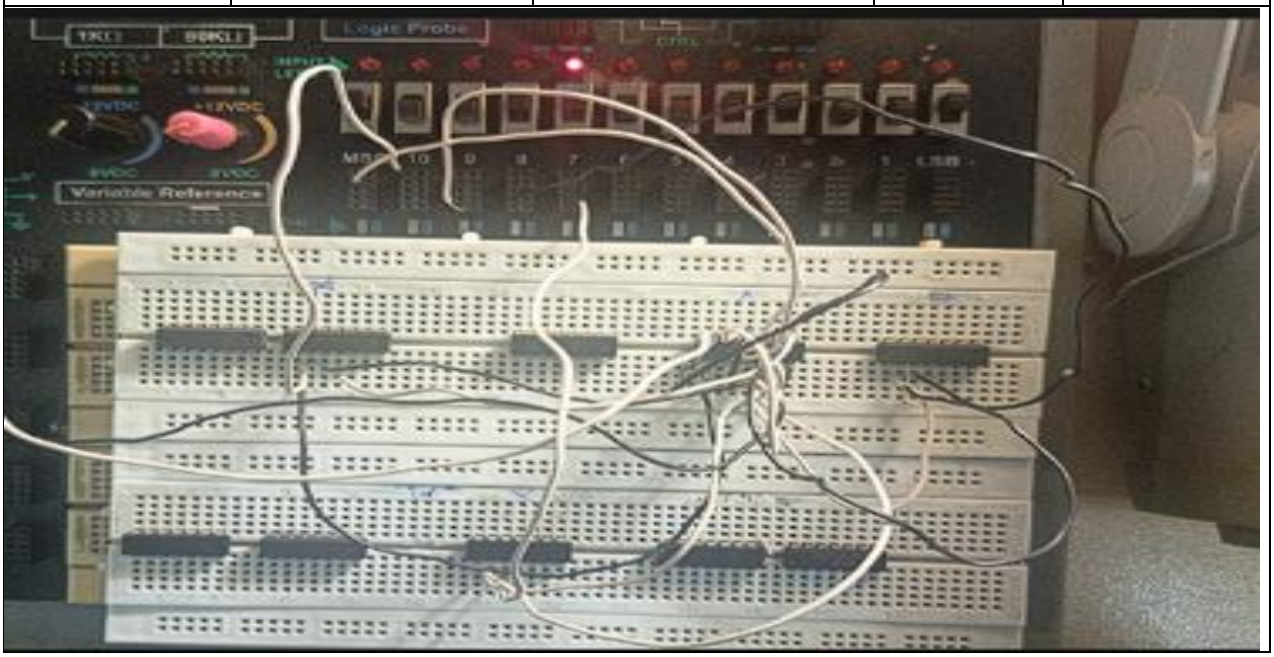
Truth Table(HALF ADDER)				
Sw	**A**	**B**	**C/B**	**S/D**
0	0	1	0	1



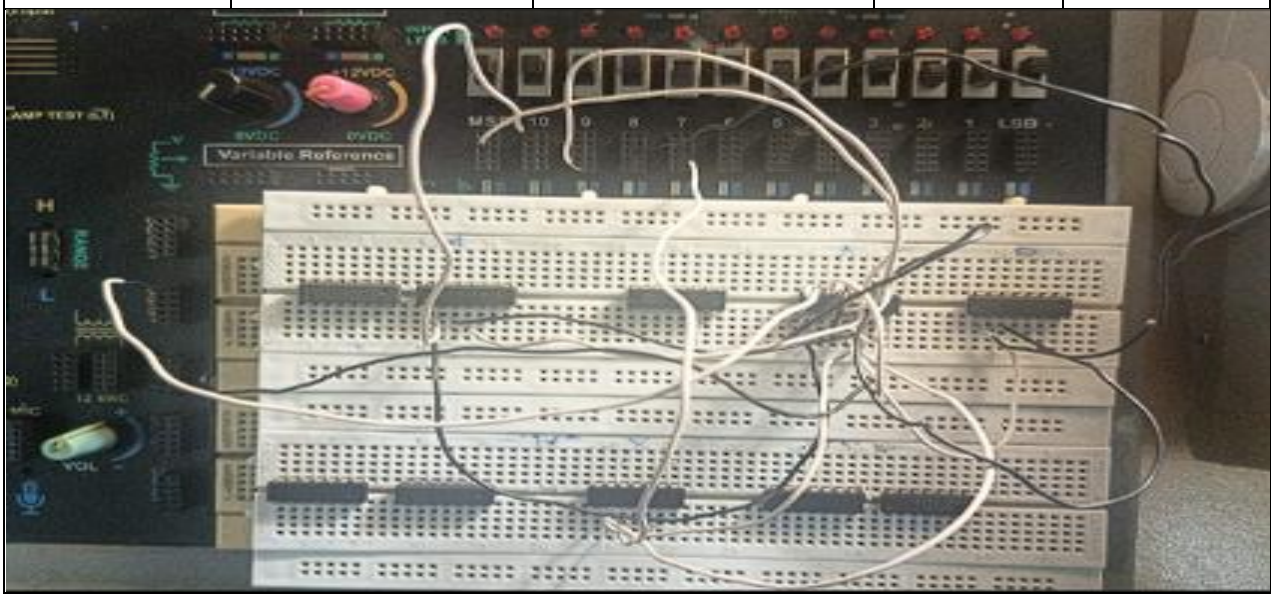
Truth Table(HALF ADDER)				
Sw	**A**	**B**	**C/B**	**S/D**
0	1	0	0	1



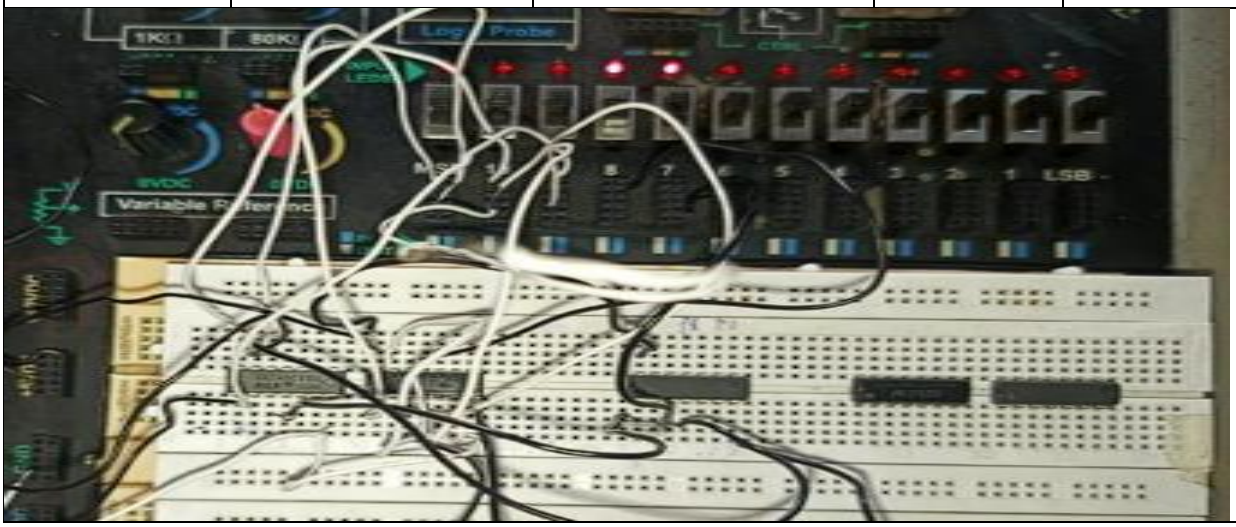
Truth Table(HALF AADDER)				
Sw	**A**	**B**	**C/B**	**S/D**
0	1	1	1	0



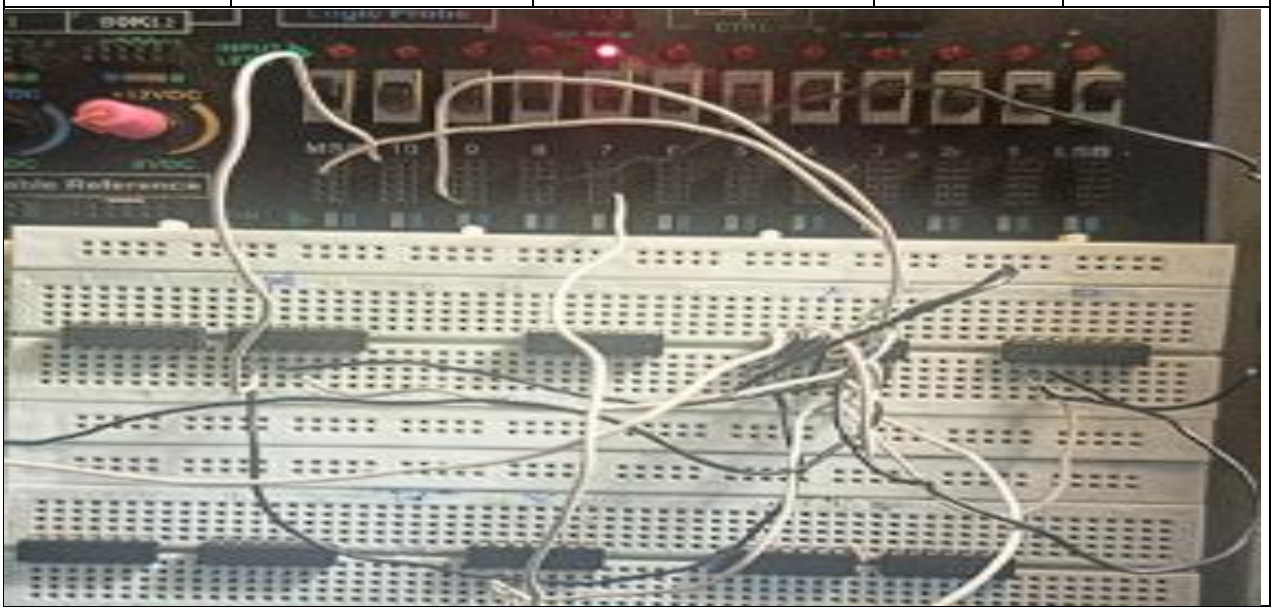
Truth Table(HALF SUBTRACTOR)				
Sw	**A**	**B**	**C/B**	**S/D**
1	0	0	0	0



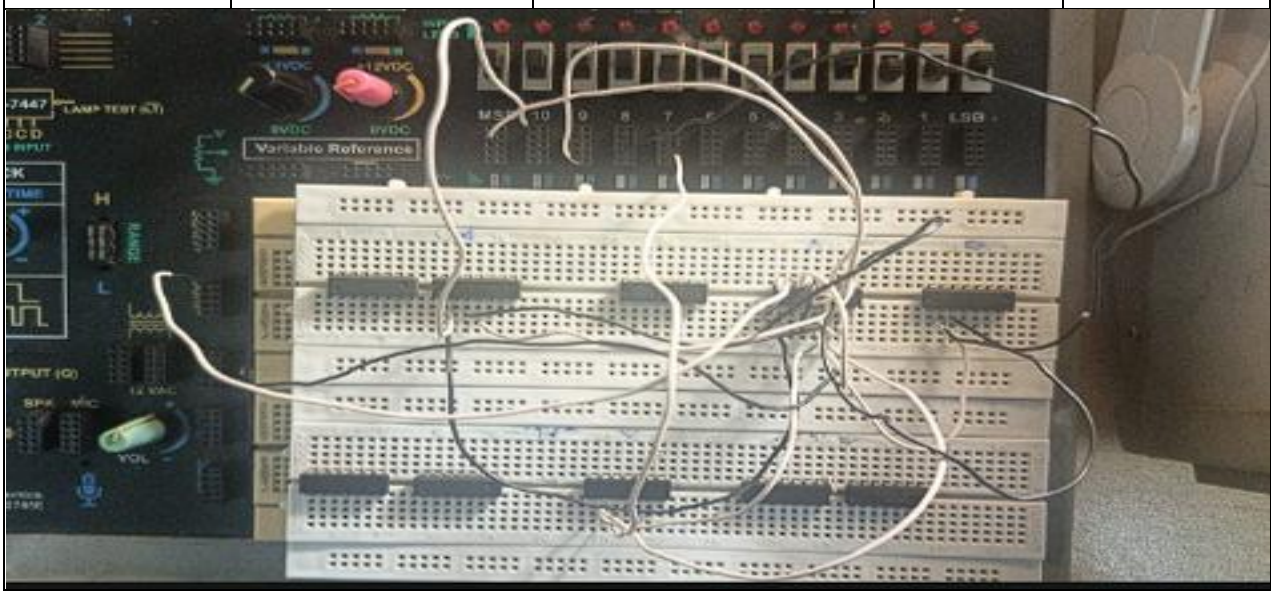
Truth Table(HALF SUBTRACTOR)				
Sw	**A**	**B**	**C/B**	**S/D**
1	0	1	1	1



Truth Table(HALF SUBTRACTOR)				
Sw	**A**	**B**	**C/B**	**S/D**
1	1	0	0	1



Truth Table(HALF SUBTRACTOR)				
Sw	**A**	**B**	**C/B**	**S/D**
1	1	1	0	0



Task 2:

Design a BCD to Seven Segment Display decoder using minimum number of logic gates.

Steps:

- Complete the truth table for a BCD to Seven Segment Decoder.
- Make K-maps for each of the output of the BCD to 7-segment decoder.
- Find out the minimal SOP for each output.
- Simplify the SOP form for each output as much as possible.
- Design the decoder using minimum number of gates.

Truth Table:

Digits	Inputs				Outputs						
	A_3	A_2	A_1	A_0	a	b	c	d	e	f	g
0	0	0	0	0	1	1	1	1	1	1	0
1	0	0	0	1	0	1	1	0	0	0	0
2	0	0	1	0	1	1	0	1	1	0	1
3	0	0	1	1	1	1	1	1	0	0	1
4	0	1	0	0	0	1	1	0	0	1	1
5	0	1	0	1	1	0	1	1	0	1	1
6	0	1	1	0	1	0	1	1	1	1	1
7	0	1	1	1	1	1	1	0	0	0	0
8	1	0	0	0	1	1	1	1	1	1	1
9	1	0	0	1	1	1	1	1	0	1	1
NOTE: Rest of the inputs are don't care											

Simplification of Logic Equations

a

AB \ CD	00	01	11	10
00	1	0	1	1
01	0	1	1	1
11	X	X	X	X
10	1	1	X	X

$a = A + C + BD + B\bar{D}$

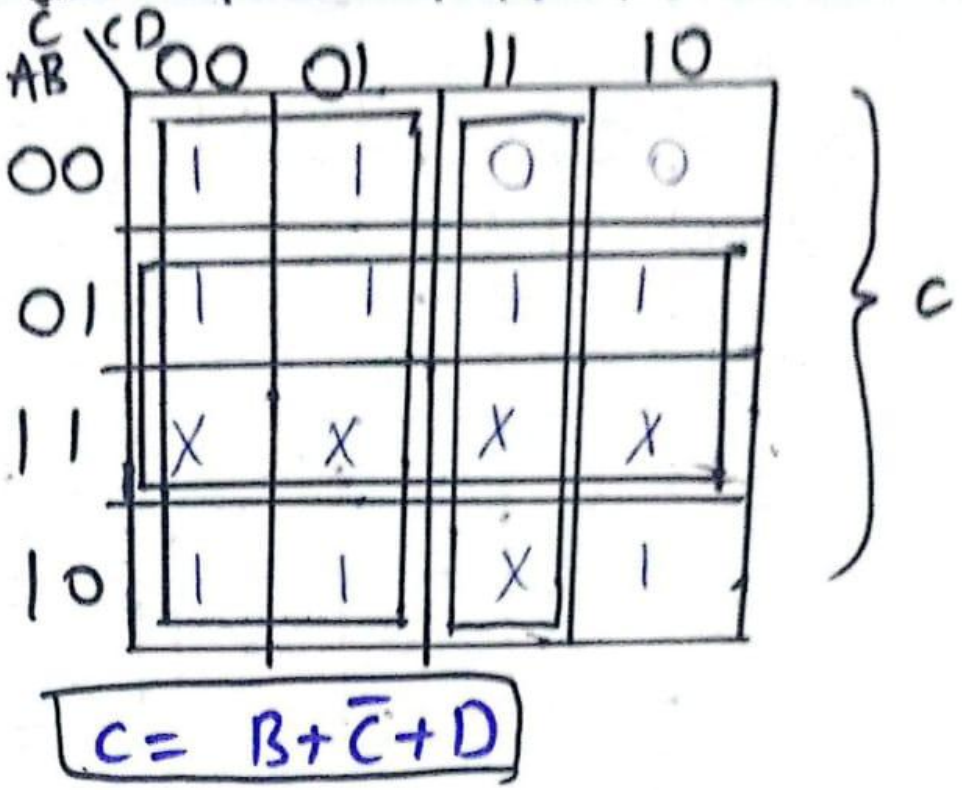
b

AB \ CD	00	01	11	10
00	1	1	1	1
01	1	0	1	0
11	X	X	X	X
10	1	1	X	X

$b = \bar{B} + \bar{C}\bar{D} + (1)$

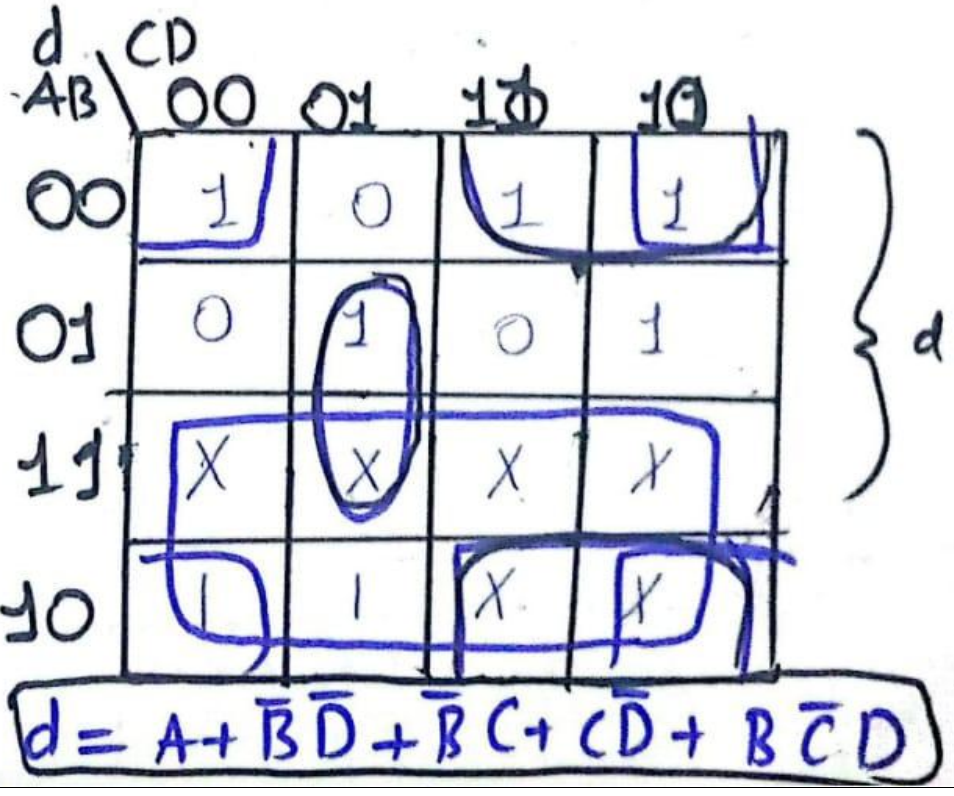
Simplification of Logic Equations

C



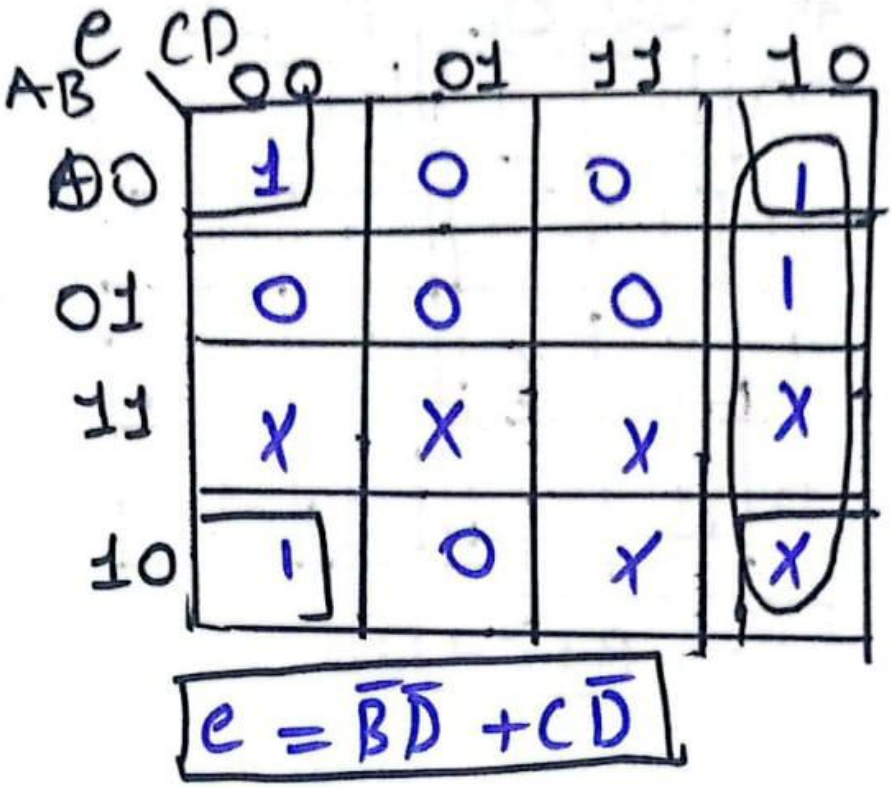
$C = B + \bar{C} + D$

D

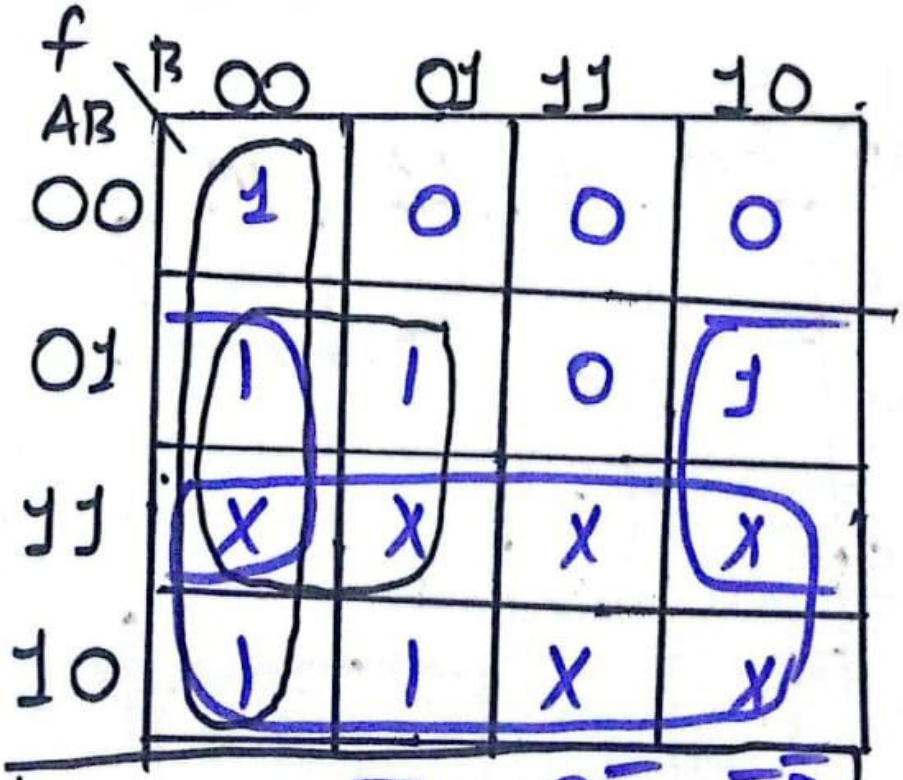


$d = A + \bar{B}\bar{D} + \bar{B}C + C\bar{D} + B\bar{C}D$

Simplification of Logic Equations

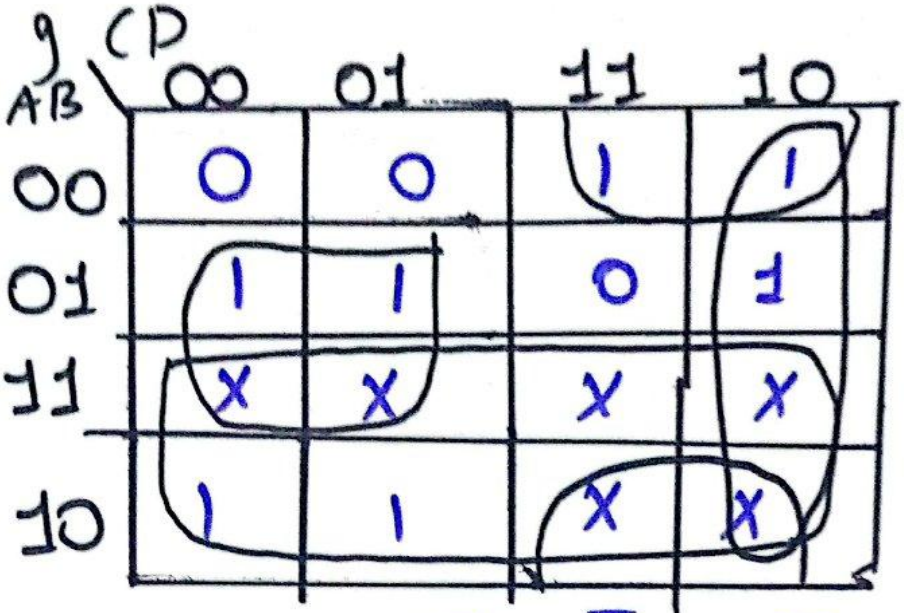


$$e = \bar{B}\bar{D} + C\bar{D}$$



$$f = A + \bar{B}C + B\bar{D} + \bar{C}\bar{D}$$

Simplification of Logic Equations



$g = A + B\bar{C} + \bar{B}C + C\bar{D}$

Final Simplified Equation (Most simplified)

$a = A + C + BD + \bar{B}\bar{D}$

$b = \bar{B} + \bar{C}\bar{D} + CD$

$c = B + \bar{C} + D$

$d = A + \bar{B}\bar{D} + \bar{B}C + C\bar{D} + B\bar{C}D$

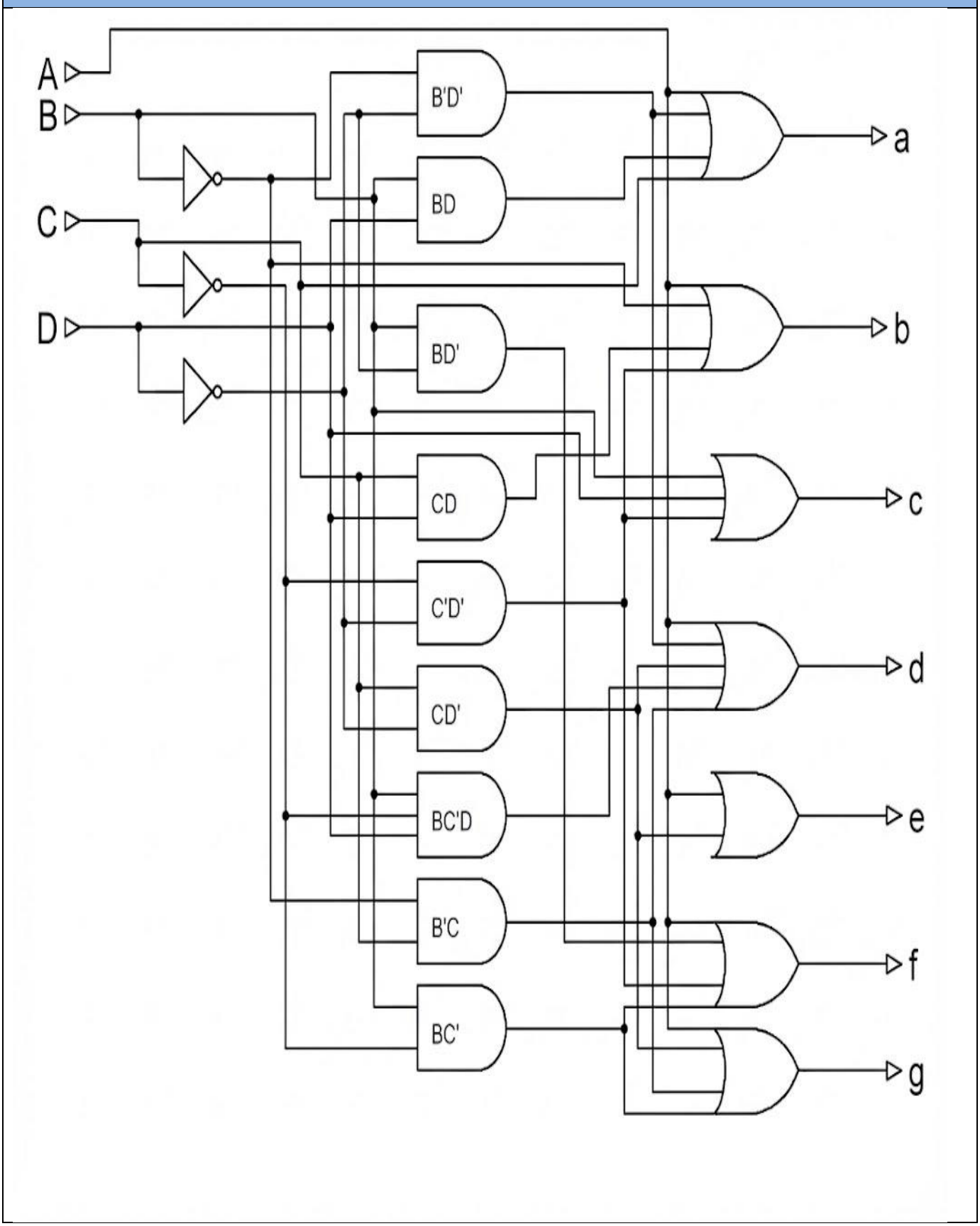
$e = \bar{B}\bar{D} + C\bar{D}$

$f = A + B\bar{C} + B\bar{D} + \bar{C}\bar{D}$

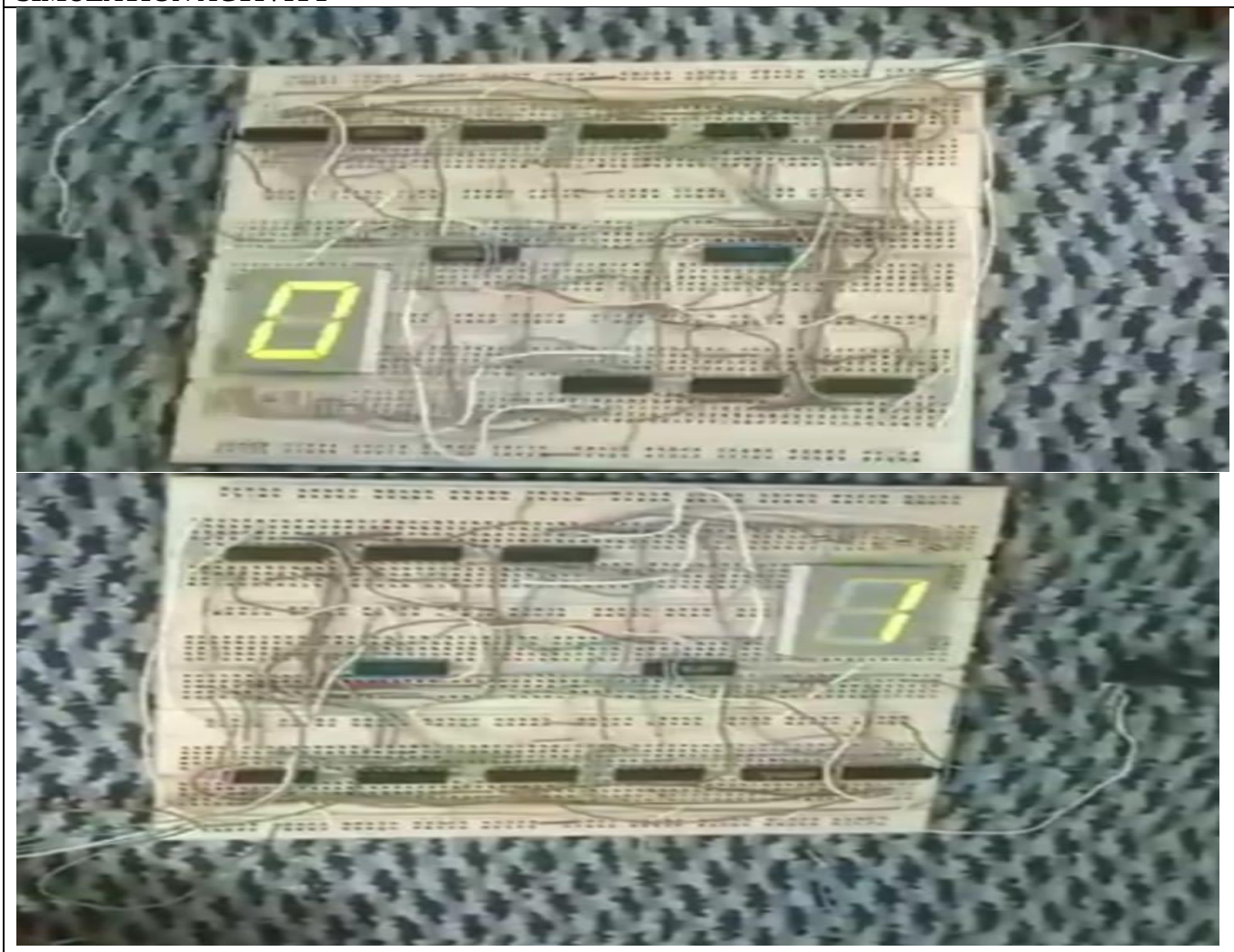
$y = A + B\bar{C} + \bar{B}C + C\bar{D}$

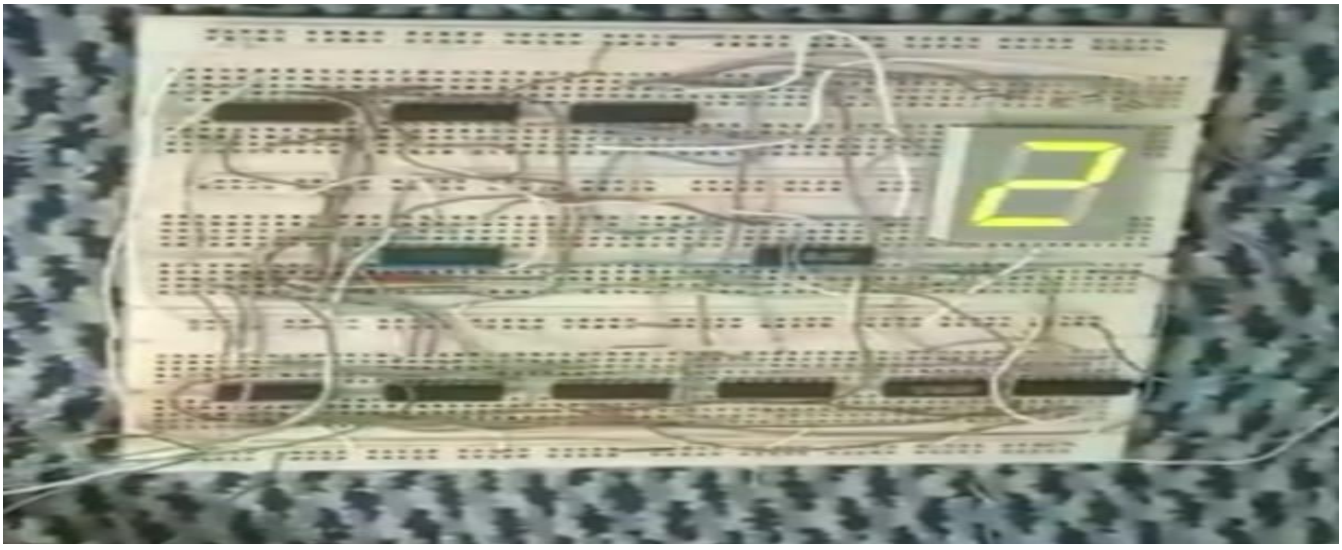
$\hookrightarrow A + C\bar{D} + (B \oplus C)$

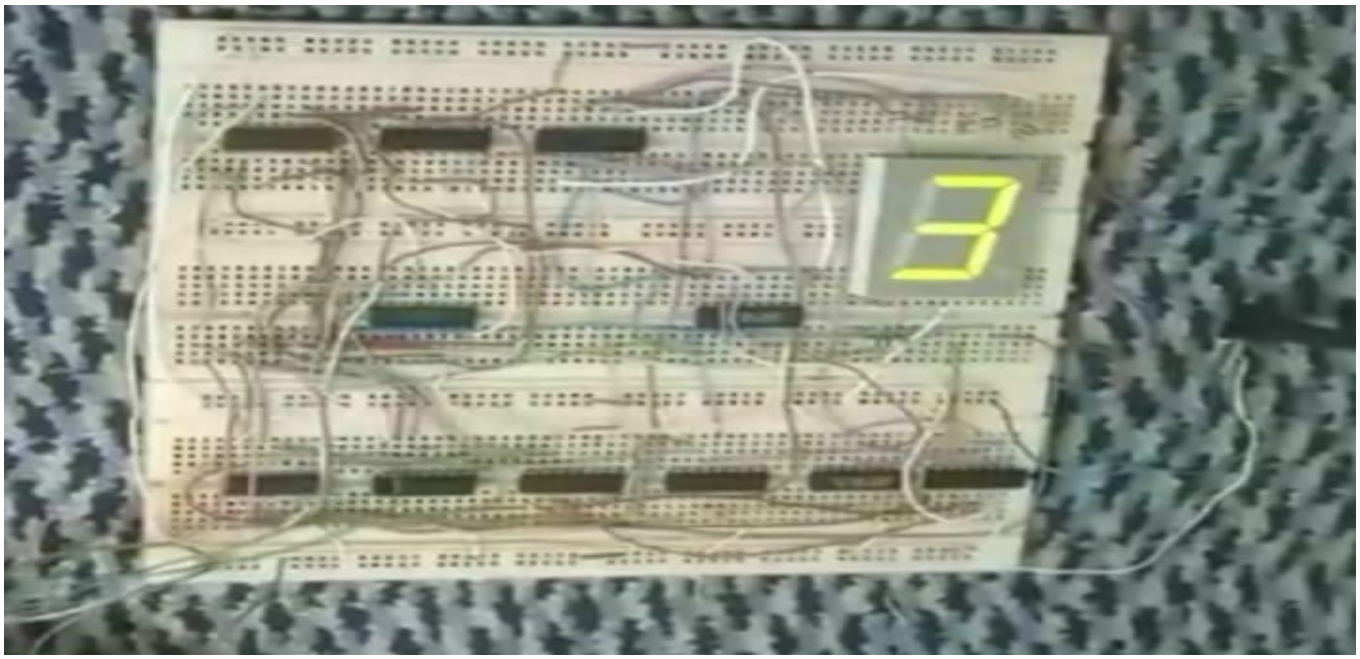
Logic Diagram:

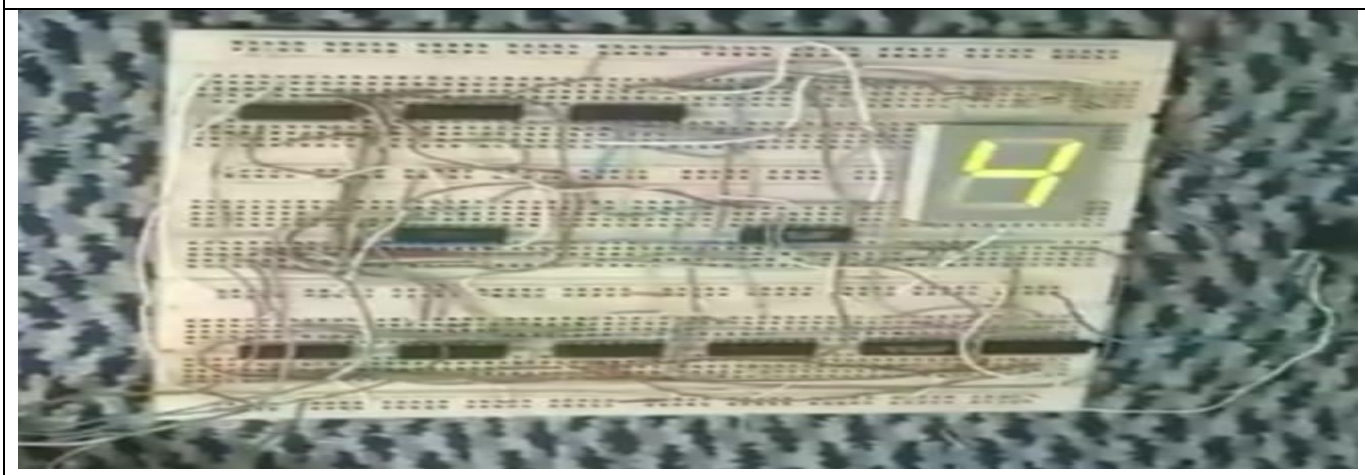


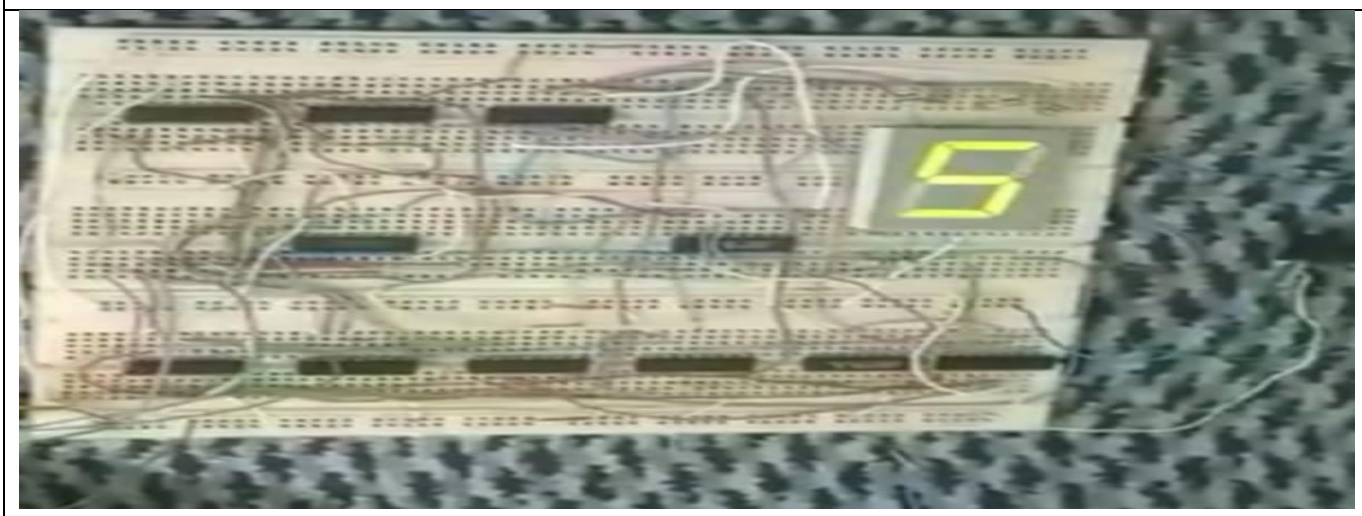
SIMULATION ACTIVITY

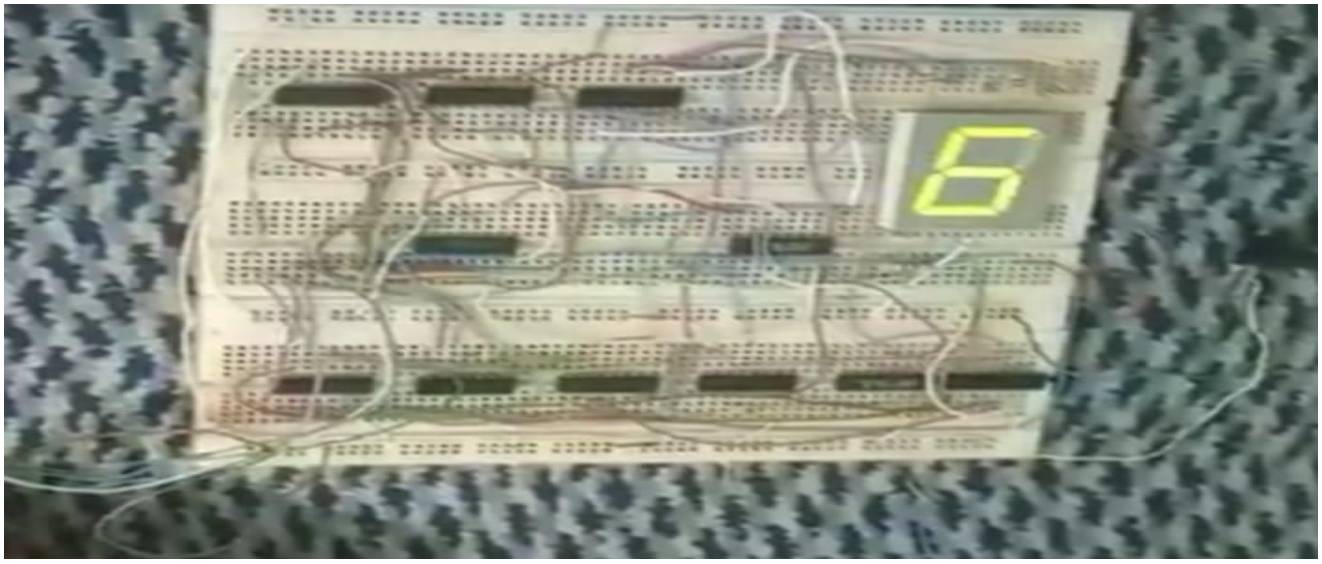


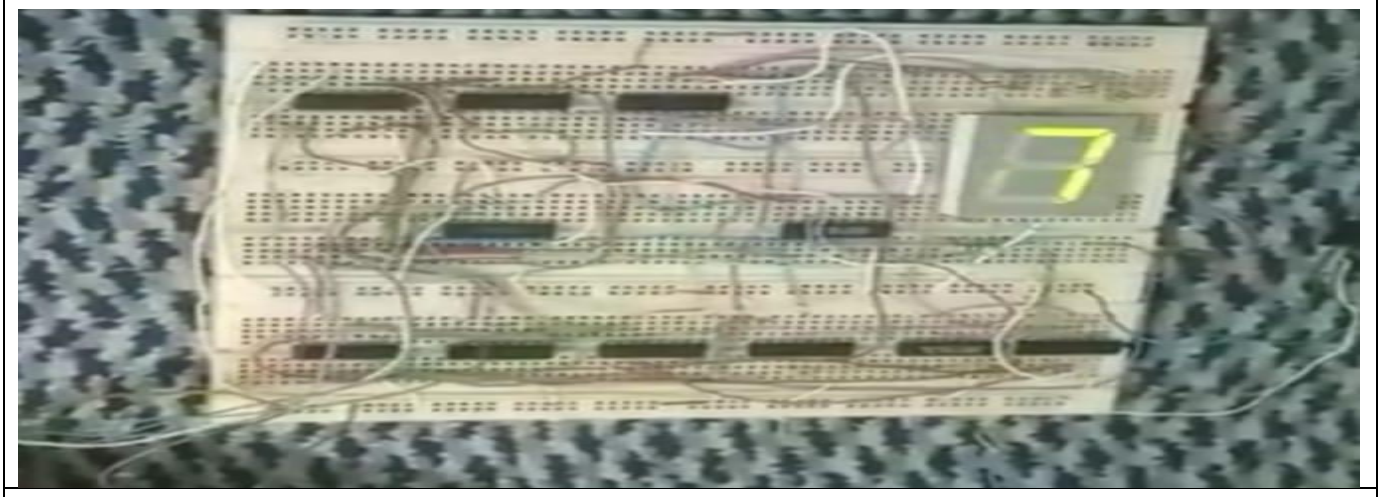


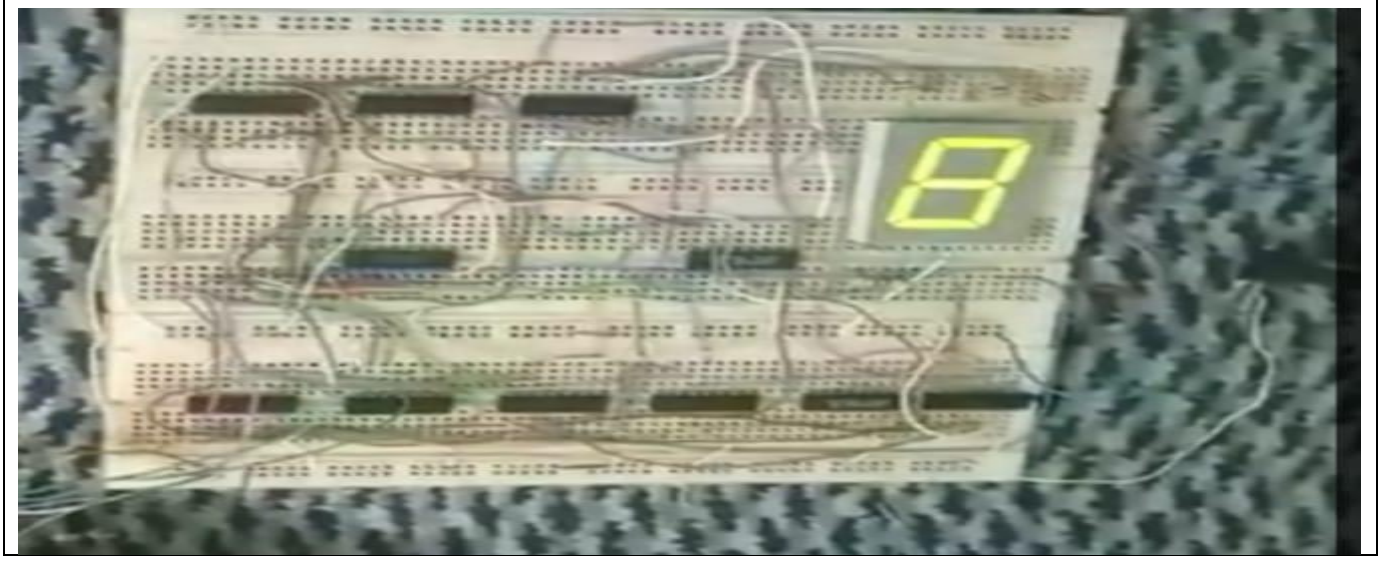

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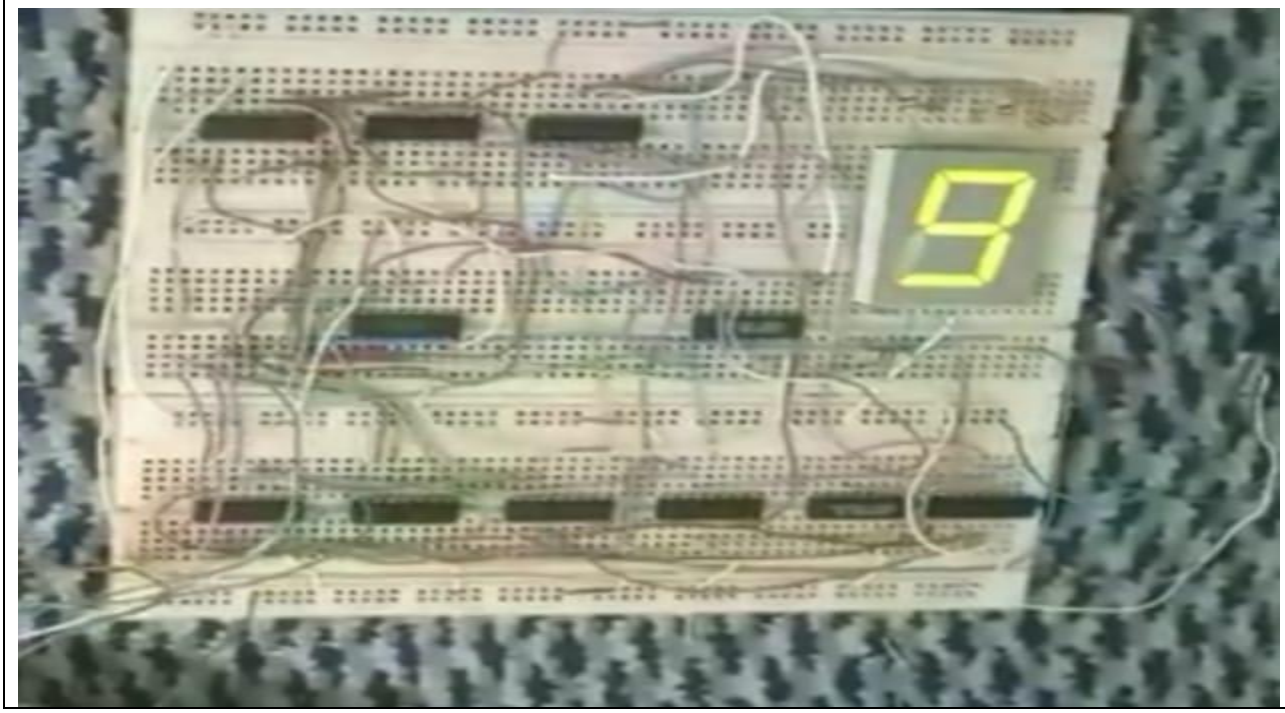

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Conclusion:

In this lab, we successfully designed and implemented two essential digital systems, demonstrating both versatility and efficiency in logic design. First, we constructed a combined half adder and half subtractor, proving that a single selection switch can control circuit behavior to perform different arithmetic operations using shared gates. Second, we engineered a BCD to 7-segment decoder by strictly following the digital design flow—from Truth Tables to Karnaugh map optimization. This process allowed us to derive the simplest Boolean expressions, enabling us to build the decoder with the minimum possible number of logic gates. Ultimately, this experiment highlighted how mathematical simplification directly translates to better hardware performance and reduced circuit complexity, bridging the gap between binary data and human-readable output.